

A Tiered Deployment Pattern for Reader Privacy: Primary Cryptographic Path, Budgeted DHT Fallback (Series Synthesis)

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Abstract

Several papers in this archive quantify a recurring empirical/operational lesson: in low-latency DHT lookups, repeated use drives per-lookup identification budgets into the 10^{-2} – 10^{-4} bit range, making dummy cover and equalization alone economically collapse (dummy fractions $w/K \rightarrow 1$) unless observation probability is extremely small. This note makes the deployment implication explicit and non-redundant: treat the raw DHT as a *budgeted fallback*, and put strong privacy mechanisms in a *primary path* whose surface is small enough for cryptographic query privacy and stricter change control.

Scope. This is an architectural synthesis note. It does not introduce new bounds; it composes the Boss-Fight dial sheet, destination-equalization contracts, and release/certification machinery into a single deployment pattern.

Imports. We treat deployed control-plane surfaces (AutoConf, delegated-routing endpoints, cache layers) as an imported state-surface checklist; see Synthesis 30.[13] Boss-Fight repeated-use budgeting (Boss Fight series) and the “dummy-cover pressure” calculation are imported.[1] Destination-privacy equalization as a receipt line item is imported from MC-EQ/CoverSketch.[2] Packaging and publication of claim inputs is imported from ABOM/OINL and release receipts.[3, 4] Threat-window declarations and trace tiers are imported from the synthesis checklists.[6, 7] Observation attenuation and scalar p_{obs} semantics are centralized in Synthesis 11; conservative combination rules for multiple witness tiers are centralized in Synthesis 15.[9, 10]

Exports. A concrete, auditable *two-path* pattern (primary + fallback) and a checklist of what must be declared in receipts so the pattern is honest.

1 Why a two-path pattern

Boss Fight budgeting (repetition) is not a “corner case”: deployed clients make many lookups per window. In the simplest “forced destination + dummies” slice, once the per-lookup budget b becomes tiny, the required dummy fraction approaches $w/K \approx p_{\text{obs}}/(2^b - 1 + p_{\text{obs}})$ (Boss Fight B).[1] Under an IPFS-style Kademlia slice and a modest corruption fraction, this pushes w/K close to 1 at $Q \approx 10^3$ lookups per window.[1]

The design implication is not that all anonymity is hopeless, but that *mechanisms must change*: either (i) reduce the repeated-use budget pressure (rate-limit, cache, coarse bucketing), (ii) reduce

observation probability dramatically (shield contacts; see Synthesis 11), and/or (iii) use cryptographic query privacy on a surface small enough to afford it.

2 Pattern: primary cryptographic path + budgeted DHT fallback

Primary path (do privacy work here). Route the majority of “routine” queries through a small, managed surface (e.g., delegated routing / indexer-like services) where cryptographic query privacy, auditing, and hardening are tractable. The primary path should export:

- a clear query-privacy mechanism description (what is hidden from whom, under what assumptions),
- any destination/privacy equalization contract (MC-EQ) if transcript structure remains observable,[2]
- change-control gates and recertification triggers for drift.[5]

Fallback path (treat raw DHT as residual). Use the raw DHT only when the primary path cannot answer (long tail, outages). The fallback path is not “free”: publish its Boss-Fight budget assumptions (window Q , observation model, and resulting per-lookup budget) as a receipt line item.[12, 1] If a deployment cannot make the fallback budget small enough, the receipt must say so.

Rate limiting as a first-class privacy knob. The most effective lever in the fallback path is the query budget Q per window. A deployment should expose Q as an explicit policy parameter (not just an abuse-control parameter) and include it in ABOM/OINL and release receipts.[12, 3, 4]

2.1 Accounting across paths (mixture bound)

If path choice is public (or independent of the secret), the overall observation channel is a mixture. A conservative bound for maximal leakage is convenient for receipts.

Theorem 1 (Mixtures of channels are log-sum-exp bounded). *Let W be a random path selector independent of the secret C , with $\Pr[W = 1] = \pi$. Let $\mathcal{L}_1$ and $\mathcal{L}_0$ be the induced transcript channels for the fallback and primary paths, with maximal leakages b_1 and b_0 . Then the mixture channel satisfies*

$$\mathcal{L}(C \rightarrow Y) \leq \log_2 (\pi 2^{b_1} + (1 - \pi) 2^{b_0}).$$

Remark 1. *This bound makes the practical requirement explicit: receipts should publish an upper bound on fallback usage π (or a monitored estimate with a confidence bound), because even a strong primary path can be dominated by frequent fallback.*

Secret-dependent fallback is worse. *If fallback selection depends on the hidden destination/profile or on secret-correlated state (cache hits, popularity, rate-limit state), then the indicator “used fallback” is itself a witness stream. In that case, mixture accounting is insufficient: one must also budget the path-selection tax and publish bounds on the secret-dependence of fallback selection (e.g., ratio bounds on π_c across declared secret pairs).[8]*

3 What to publish so the pattern is honest

ABOM/OINL and release receipts should include at least one line item per path using the series' minimal schema (Synthesis 12).[12] To keep the boundary legible, publish:

- the path policy (when each path is used) and the transcript tier(s) it exposes,
- the threat window and usage assumptions (e.g., Q per window) and a bound/estimate for fallback usage π ,
- the observation parameter(s) p_{obs} (or monitored upper bounds); if multiple tiers are published, include a tiered observation vector and its combination rule,[9, 10]
- the per-path budget value(s) and the knob hashes that determine the accountant (dummies, timeouts, equalization parameters).

4 Takeaways

- **Stop trying to make the raw DHT do everything:** use it as a fallback with an explicit, published Boss-Fight budget.
- **Put strong privacy in the primary path:** small surfaces are where cryptographic query privacy and strict change control are feasible.
- **Publish the boundary:** receipts must state when fallback happens and what it costs.

References

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